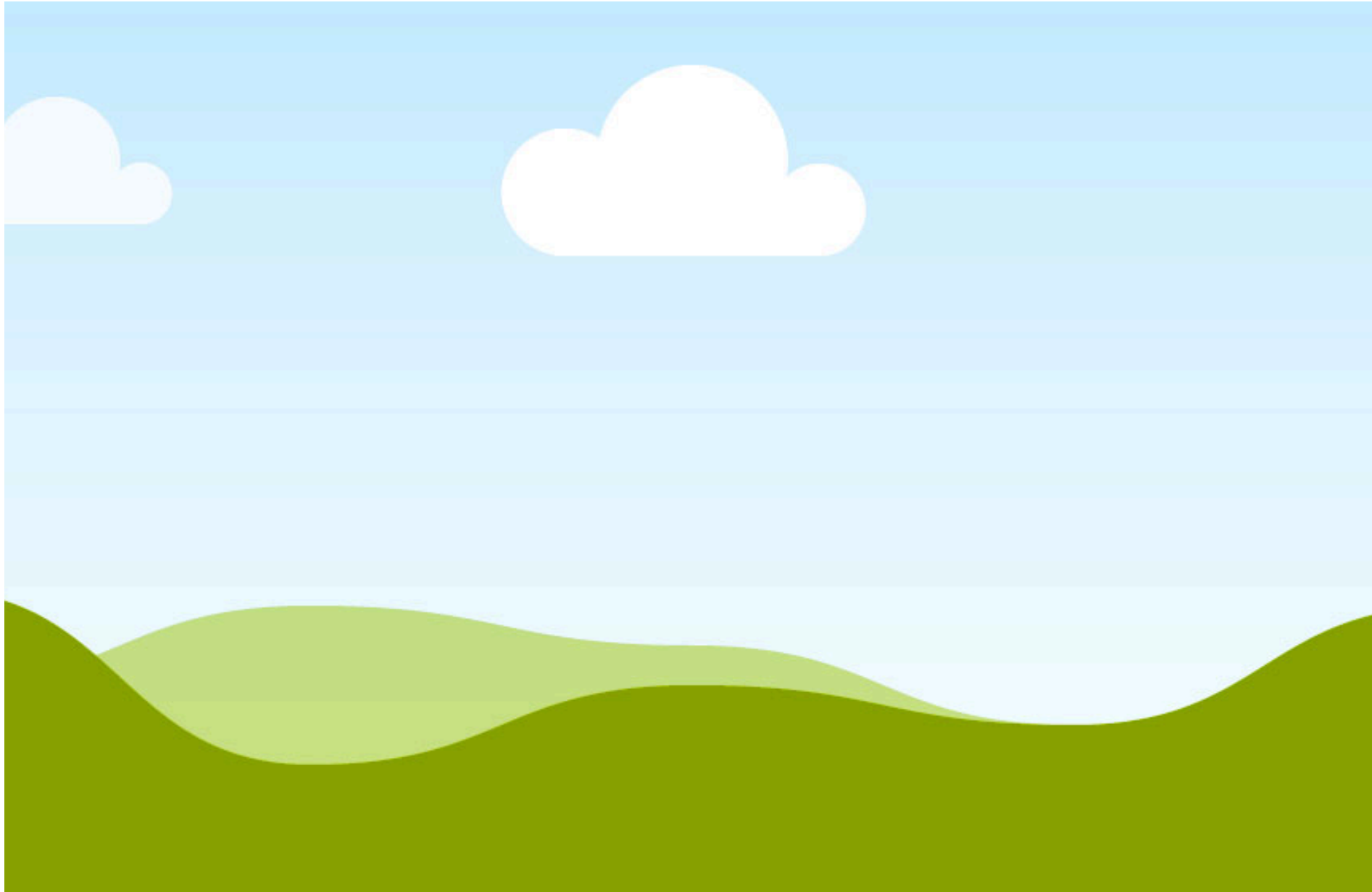


Introduction to Data Visualization

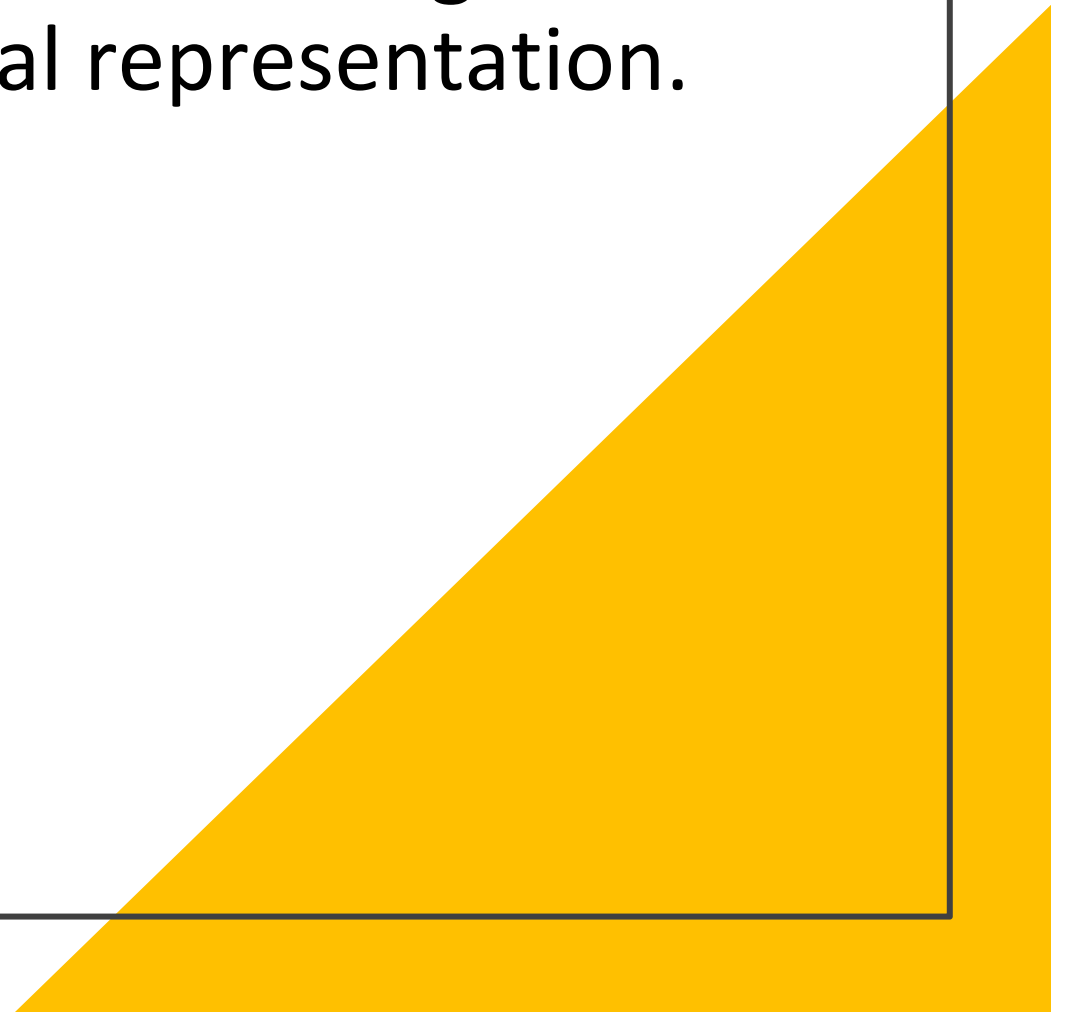
H. M. Samadhi Chathuranga Rathnayake
MSc, MBA, PGDip, BSc (1st Class)





Data Visualization

- Process of converting data into a graphical representation.



Data Visualization

- Type of variables is one of the main factors to be considered when visualizing
 - Numerical variables (Ex- Age, Height, Weight)
 - Categorical variables (Ex- Gender, Color, District, Country)

Line Chart

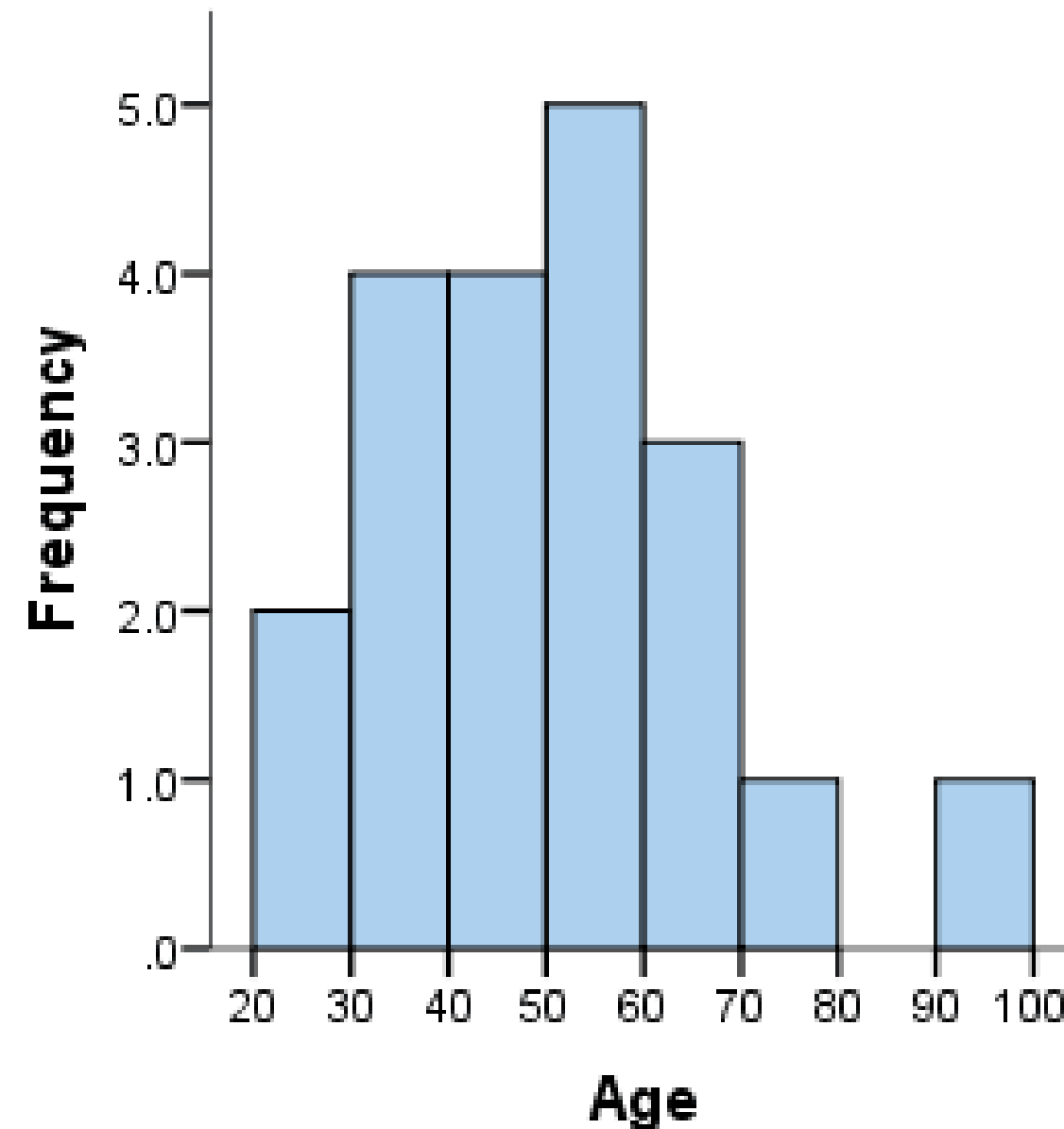
- A line chart or line plot or line graph or curve chart is a type of chart which displays information as a series of data points called 'markers' connected by straight line segments.

ZZD to QQY Exchange Rates



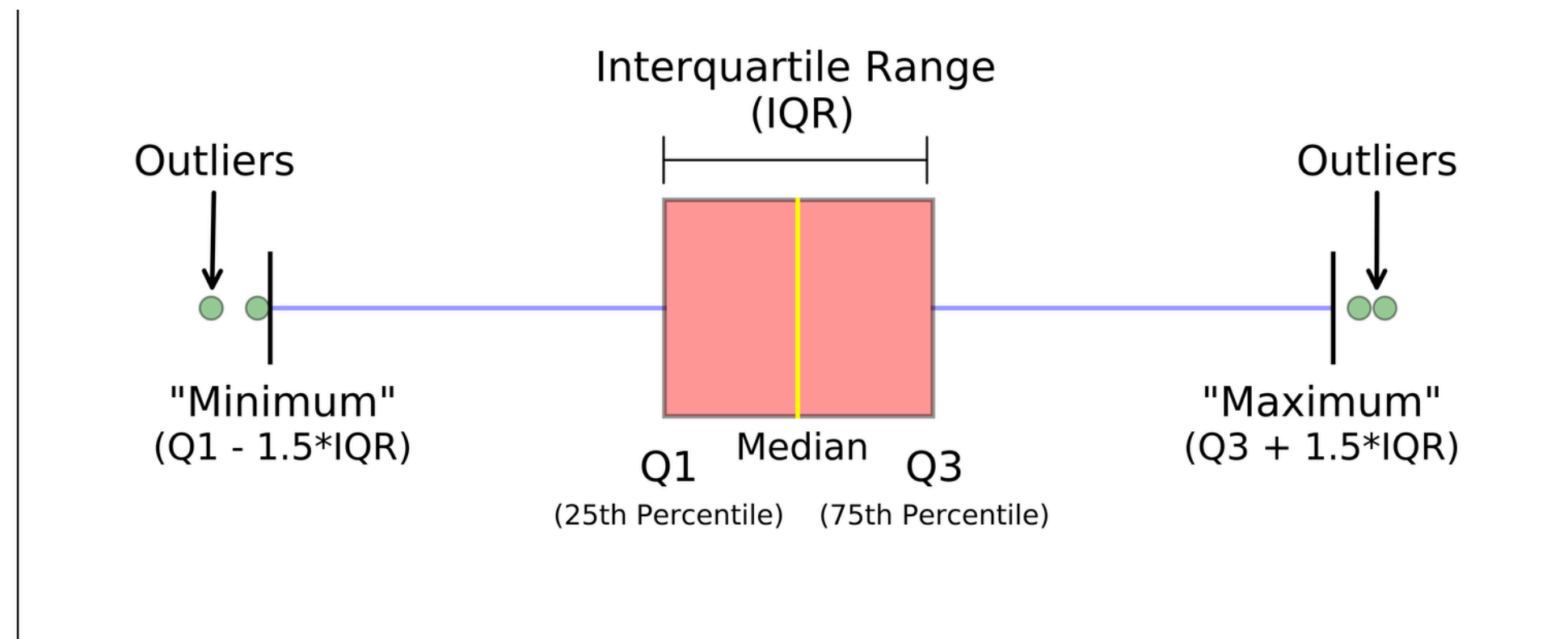
Histogram

- A histogram is an approximate representation of the distribution of numerical data.



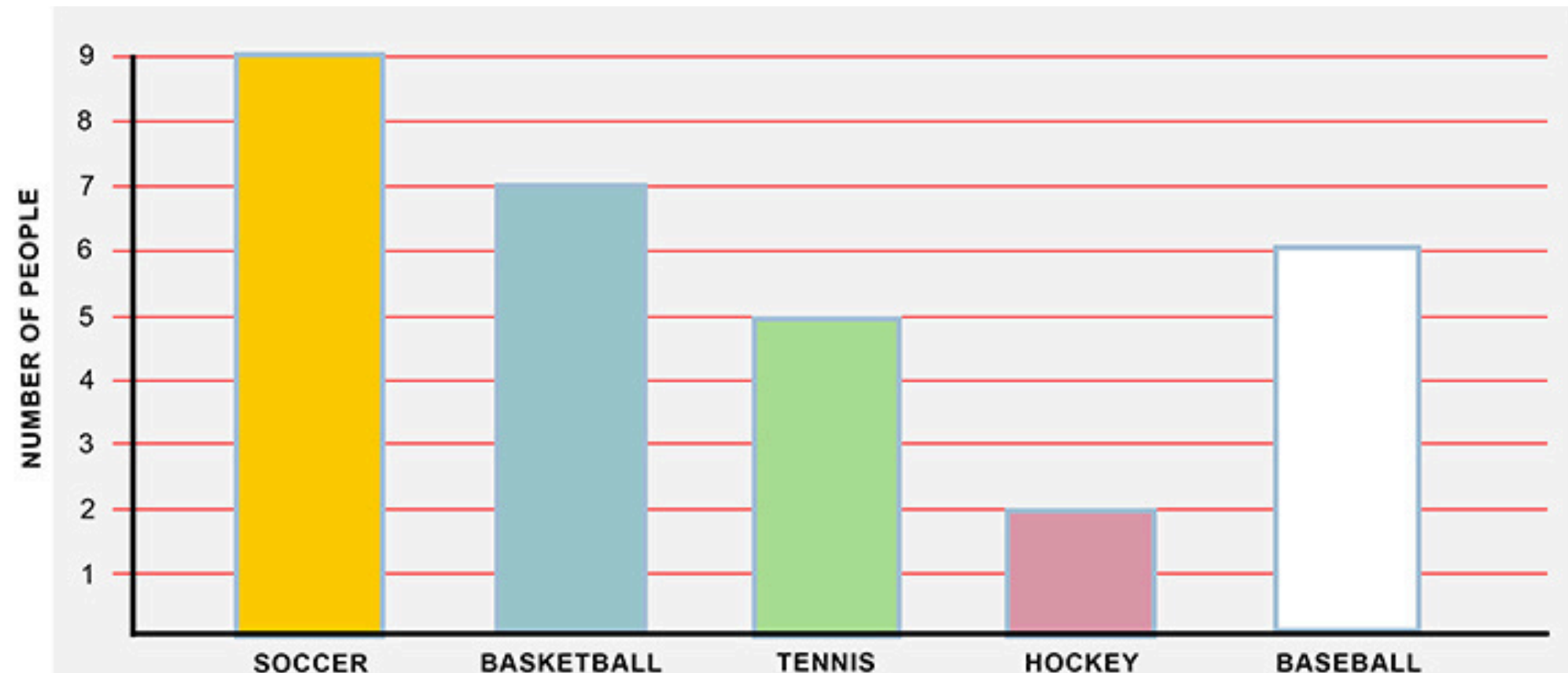
Boxplot

- In descriptive statistics, a box plot or boxplot is a method for graphically depicting groups of numerical data through their quartiles (5 Number Summary).
- They are showing outliers as well.



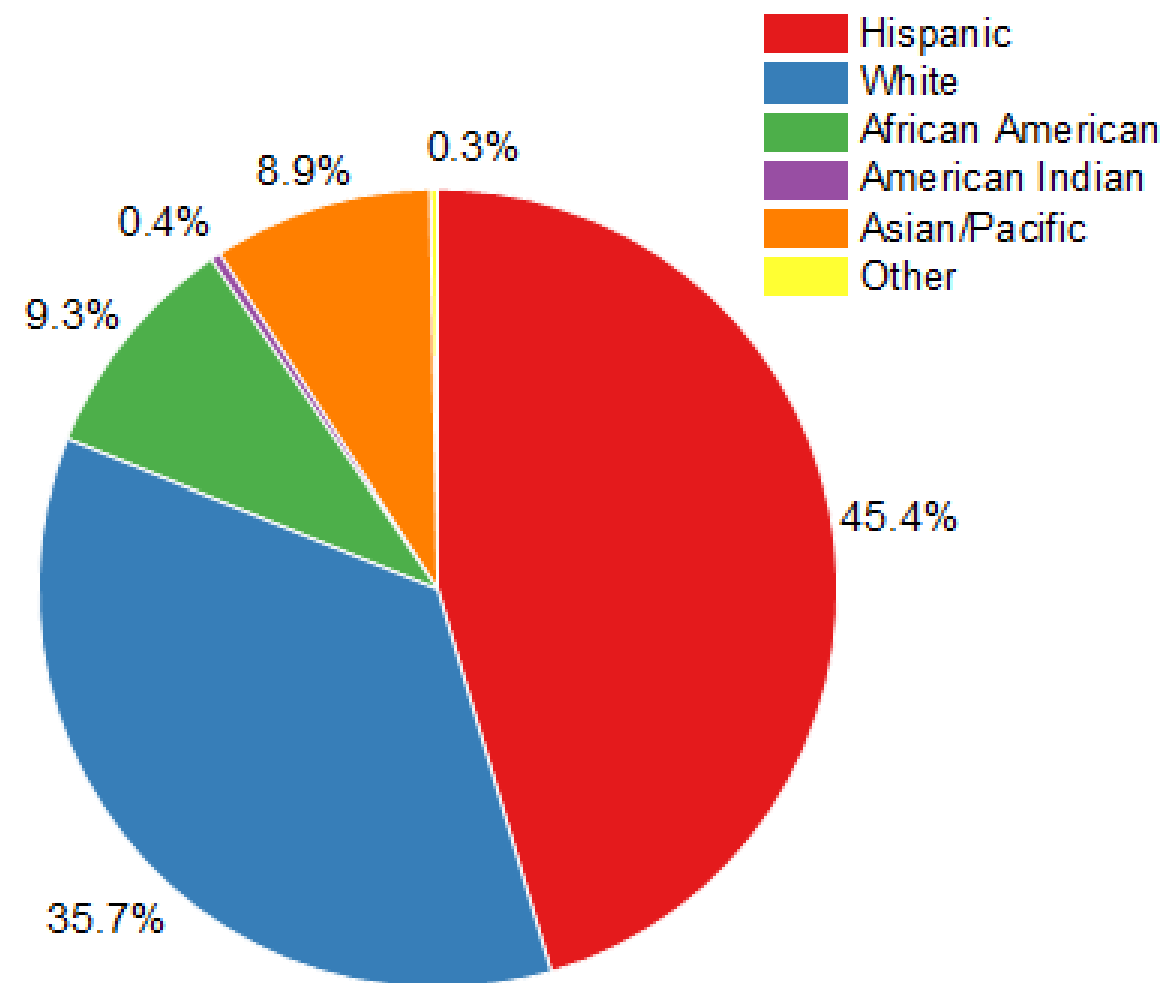
Bar Chart

- A bar chart or bar graph is a chart or graph that presents categorical data with rectangular bars with heights or lengths proportional to the values that they represent.
- The bars can be plotted vertically or horizontally.



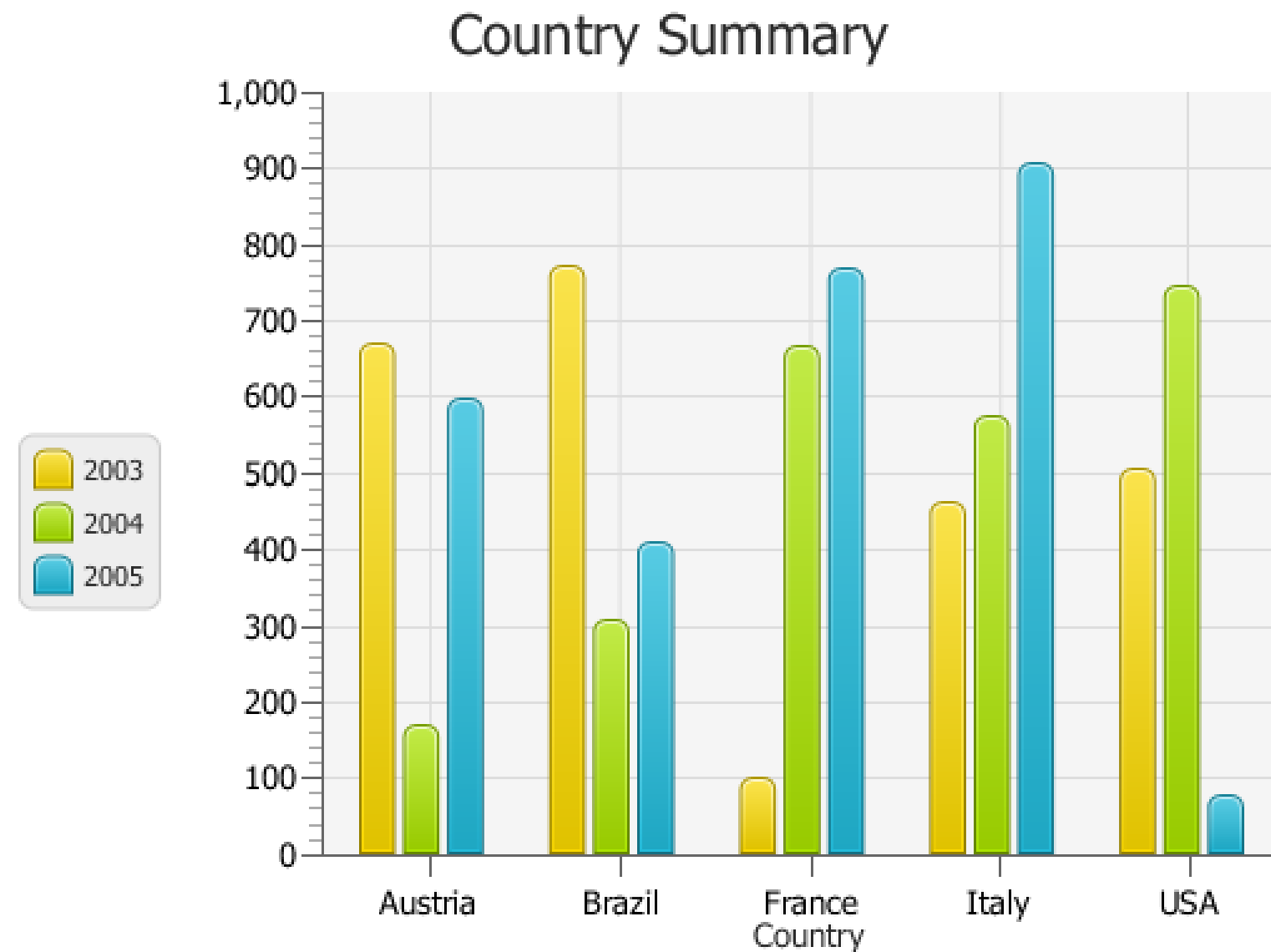
Pie Chart

- A pie chart is a circular statistical graphic, which is divided into slices to illustrate numerical proportion.



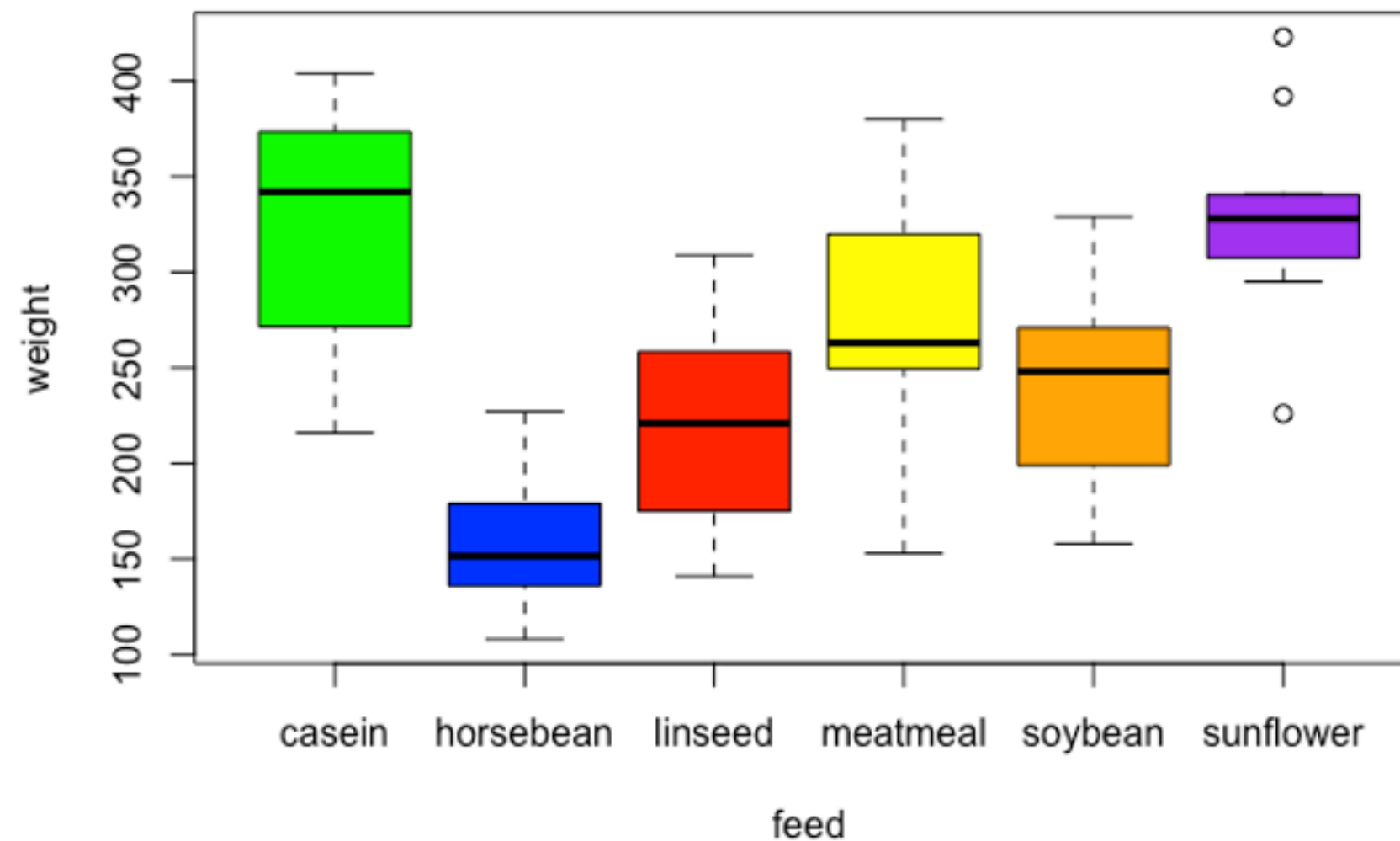
Multiple Bar Chart

- Side by side multiple bar charts show how two or more categories are related



Side by Side Boxplots

- Side-By-Side boxplots are used to display the distribution of several quantitative variables or a single quantitative variable along with a categorical variable.



Scatter Plots

- A scatter plot is a type of plot or mathematical diagram using Cartesian coordinates to display values for typically two variables for a set of data.

